The Ghost in the Cache: Forensic Carving of 'Deleted' LLM Chats and Unsubmitted Prompts from Chromium Storage

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Abstract

When users click 'Delete Chat' inside ChatGPT, Claude, or Gemini interfaces, cloud records are marked for removal, but local Chromium LevelDB storage retains write-ahead log records (.log), uncompacted SSTables (.ldb), and unencrypted blob files (.indexeddb.blob/). Across 23 browser profiles, we recovered 506 text artifacts, 70.8% of which had already been deleted from the UI. We also uncover pre-submission keystroke draft persistence in Claude TipTap, unencrypted 10MB PDF document blob persistence, and raw WebRTC ECDSA private keys in ChatGPT Desktop. We introduce **Tinker Tailor LLM Spy**, a zero-dependency, sub-163ms forensic framework. Official repository: https://github.com/sapnadfir-blip/ghost-in-the-cache

1. Introduction & The Ephemeral AI Fallacy

Client-side Generative AI telemetry is a critical blind spot in endpoint security. Because LevelDB is an append-only LSM tree, deleted records persist until compaction. Our lock-free carver reconstructs conversations, keystroke drafts, and uploaded document attachments directly from raw binary streams.

2. Critical Vulnerability Discoveries (13 Classes)

• **LevelDB LSM Remanence:** 70.8% of recovered chat artifacts were UI-deleted, persisting up to 83 days.
• **TipTap Keystroke Drafts:** In-flight prompt drafts are cached before clicking Send.
• **Plaintext Document Blobs:** Uploaded PDFs and code files persist unencrypted in blob storage.
• **Desktop App Key Exposure:** ChatGPT Desktop persists ECDSA private keys in LevelDB WAL logs.

3. Defensive Mitigations & Tool Release

Mitigations include GPO cache eviction policies, BitLocker/FileVault encryption, and EDR path monitoring. The open-source forensic tool, whitepaper, and reproduction suites are available at: https://github.com/sapnadfir-blip/ghost-in-the-cache